

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First / Second Semester B.Tech. (CBCS) Degree Examinations

Summer Semester – July 2024

EE1001- 1 – BASIC ELECTRICAL ENGINEERING

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Peak factor of a sinusoidal current is _____.
 A) 0.707
 B) 1.414
 C) 1.11
 D) None of these
- If an alternating current has its RMS value 5 A, frequency 60 Hz, its instantaneous value is given by
 A) $i = 5 \sin 377t$ A
 B) $i = 5/\sqrt{2} \sin 377t$ A
 C) $i = 5\sqrt{2} \sin 377t$ A
 D) None of these
- According to KCL as applied to junction in a network _____.
 A) Total sum of currents meeting at the junction is zero
 B) No current can leave the junction without some current entering it
 C) Net current flow at the junction is positive
 D) Algebraic sum of current meeting at the junction is zero
- The power factor of an AC circuit is given by _____.
 A) Cosine of the phase angle
 B) Tangent of the phase angle
 C) The ratio of R by X_L
 D) The ratio of X_L by Z
- If a 100V, 50Hz, single phase AC supplies a current of 2A to a pure capacitive circuit, the capacitance of the circuit is _____.
 A) 50 F
 B) 637mF
 C) 63.7 μ F
 D) 0.159F
- The capacitive reactance is measured in _____.
 A) ohm
 B) farad
 C) henry
 D) None of these
- A 1.2 k Ω resistor is connected in series with a 15mH inductor and energized by a 10V, 10 kHz, 1- ϕ supply. Then, the circuit impedance is _____.
 A) 1526 Ω
 B) 152.6 Ω
 C) 1200 Ω
 D) 942 Ω
- In a certain series circuit, if the impedance is $(4-j6) \Omega$, it means that _____.
 A) $R = 4 \Omega$; $X_L = 6 \Omega$
 B) $R = 6 \Omega$; $X_L = 4 \Omega$
 C) $R = 4 \Omega$; $X_C = 6 \Omega$
 D) $R = 6 \Omega$; $X_C = 4 \Omega$
- The secondary voltage of a 10kVA transformer with load current of 10A is _____.
 A) 10kV
 B) 100kV
 C) 1kV
 D) none of the above
- In a given transformer for given applied voltage, which of the following losses remain constant irrespective of load changes?
 A) Friction and windage losses
 B) Copper losses
 C) Hysteresis and eddy current losses
 D) Cannot be determined
- Strength of an electromagnet can be increased by _____.
 A) Increasing the cross-sectional area
 B) Increasing the number of turns
 C) Increasing current supply
 D) All of the above

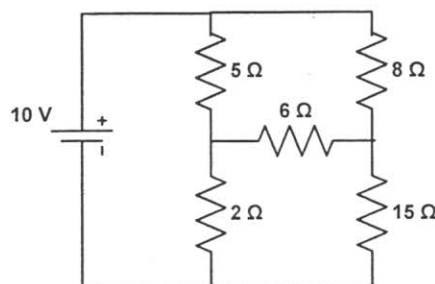
12. The magnetic reluctance of a material _____
 A) Decreases with increasing cross-sectional area of material
 B) Increases with the increasing cross-sectional area of material
 C) Does not vary with the increasing cross-sectional area of material
 D) None of these
13. The material used in brushes of DC generator is
 A) Carbon
 B) Copper
 C) Both (a) and (b)
 D) None of the above
14. The number of parallel paths in 8 pole wave wound DC generator is
 A) 2
 B) 16
 C) 8
 D) 4
15. A machine without commutator, providing an ac emf to the external circuit is called as
 A) D.C. generator
 B) Alternator
 C) Synchronous motor
 D) Transformer
16. The power factor of an alternator depends on
 A) Load
 B) Speed of rotor
 C) Core losses
 D) Armature losses.
17. In a three phase induction motor, the synchronous speed of the rotating magnetic field is
 A) $N_s = 120 f / P$
 B) $N_s = 60 f / P$
 C) $N_s = 120 P / f$
 D) $N_s = 120 f P$
18. The difference between the synchronous speed N_s of the magnetic field and the actual speed of the rotor N is called as the
 A) Synchronous speed
 B) slip speed
 C) Asynchronous speed
 D) maximum speed
19. The earth wire should be
 A) good conductor of electricity
 B) mechanically strong
 C) both (a) and (b)
 D) mechanically strong but bad conductor of electricity.
20. Resetting is quick and simple in
 A) switch
 B) MCB
 C) Fuse
 D) none of these.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Using mesh analysis, determine the value of voltage across $6\ \Omega$ resistor for the circuit shown in FigQ1a.



FigQ1a

- b) Prove the relationship between line and phase quantities in a delta connected 3ϕ system with the help of phasor diagram. Also derive the equation of power in a 3 phase system in terms of line parameters.
2. a) Define average value of an alternating sinusoidal waveform. Also derive the expression for the same.

8	L3	1	1,2
8	L4	2	1,2
8	L3	1	1,2

- b) Three coils each having a resistance and inductance of 8Ω and $0.02H$ respectively are connected in star across a three phase 230 V , 50 Hz supply. Find the impedance, line current, power factor and power consumed by the load.
- 8 L3 2 1,2
3. a) Prove that power consumed by pure capacitive circuit is zero.
- 8 L1 1 1,2
- b) Two impedances $Z_1 = (10 + j15)\Omega$ and $Z_2 = (6 - j8)\Omega$ are connected in parallel. If the total current of the combination is $15A$, what is the power taken by each branch?
- 8 L3 2 1,2
- Unit – II**
4. a) With neat diagram explain the construction of DC generator.
- 8 L1 4 1,2
- b) In a 50 KVA , $11000/220\text{ V}$ single phase transformer the iron and full load copper losses are 350 W & 425 W respectively. Find (i) Efficiency at full load unity power factor (ii) The load at which maximum efficiency occurs (iii) Max efficiency.
- 8 L3 3 1,2
5. a) Elaborate on following terminology.
- (i) Faraday's Laws
- (ii) Self-Inductance
- (iii) Mutual Inductance
- 4 L1 3 1,2
- b) Derive an expression for the EMF induced in transformer.
- 6 L1 3 1,2
- c) A 4-pole generator has a lap-wound armature with 50 slots with 16 conductors per slot. The useful flux per pole is 30 mWb . Determine the speed at which the machine must be driven to generate an EMF of 240 V .
- 6 L3 4 1,2
6. a) Derive the condition for maximum efficiency in transformer.
- 4 L2 3 1,2
- b) A 230 V D.C. shunt motor takes 32 A at full load. Find the back EMF on full load if the resistances of armature and shunt field windings are 0.2Ω and 115Ω respectively.
- 6 L3 4 1,2
- c) A 12 pole 500 rpm star connected alternator has 48 slots with 15 conductors/slot, the flux/pole is 0.02 Wb . Assume unity distribution factor and winding factor as 0.97 . Calculate line EMF.
- 6 L3 4 1,2
- Unit – III**
7. a) What is earthing. Explain pipe earthing.
- 8 L1 5 1,2
- b) With neat table and diagram explain three-way control of lamps.
- 8 L1 5 1,2
8. a) With neat diagram explain the concept of rotating magnetic field in an induction motor.
- 8 L2 5 1,2
- b) A 3 Phase Induction Motor has 6 poles and runs at 960 RPM . on full load. It is supplied from an Alternator having 4 poles and running at 1500 RPM . Calculate the full load slip and frequency of the rotor currents of the Induction motor.
- 8 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
